

FINAL YEAR PROJECT REPORT

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DEGREE COURSE:	BSc Physics
PROJECT TITLE:	Identifying Decays of the Higgs Boson to Dark Matter at the Large Hadron Collider with Machine Learning Algorithms
YEAR OF SUBMISSION:	2026
SUPERVISOR:	Henning Flaecher
NUMBER OF WORDS:	5581



Declaration

All simulated data used in this project was provided by the project Supervisor at the start of the project alongside a GitHub repository containing skeleton code for the particle transformer pipeline developed. This code did not run at first, and needed substantial modification to achieve the results presented in this report. Project-directional guidance was provided by Supervisor Henning Flaecher and PhD student Florian Bury throughout the project timeline. The feature-level physics validation in §4.1.6 was my original idea.

Since the project was coding-based, my partner and I worked independently on this project. However, we discussed and shared ideas in weekly project group meetings including our Supervisor and PhD student. All analysis, code, and findings presented in this report was my own.

AI Declaration

As part of the research described above, Perplexity Research and Claude Sonnet were used in the research phase to (1) clarify new physics concepts (2) debug code errors when stuck (3) generate boilerplate segments of code in accordance with the Project Unit’s AI-use policy. AI tools were not used in the writing or production of this report besides for proofreading. AI-assisted code generation is documented in the accompanying notebook.

Acknowledgements

I thank Prof. Henning Flaecher for providing the Monte Carlo datasets and skeleton code, along with patient guidance throughout the project. PhD student Florian Bury offered valuable insights during our discussions. Both greatly contributed to my understanding of this project. I also thank Jeronimo Segal for writing the skeleton code used in this project. Lastly, thank you to Luke Kreczko for his help in organising and granting access to the DICE network.

Preface

Superscript footnotes have been included in this report to expand upon standard HEP/ML terminology potentially novel to non-HEP readers.

Abstract

The search for dark matter via the Higgs portal mechanism — constraining $\text{BR}(H \rightarrow \text{inv})$ — in the $t\bar{t}H$ channel has yielded the weakest limits amongst Higgs production modes (CMS: $< 43\%$ at 95% CL). Using LHC Run 2 simulations (integrated luminosity of 138fb^{-1}), this project deploys a Particle Transformer multiclassifier with MET decorrelation (DisCo loss), achieving $\text{AUC} = 0.795$. It yielded an expected upper limit of 31.3% at 95% CL on simulated events using Run 2 integrated luminosity and flat background yield uncertainty of 10%. At HL-LHC integrated luminosity of 3ab^{-1} , the model yielded an expected upper limit of 17.5% at 95% CL. The model was found to have learnt expected physics, as validated by score-feature analyses. This expected-only analysis suggests event-level ML improves sensitivity over prior methods.

1 Introduction

Astrophysical evidence establishes dark matter (DM) as a major fundamental constituent of the universe [1–3]. The Higgs Portal Mechanism — a beyond standard model (BSM) framework that is agnostic to the spin of DM particles [4] — is a well-motivated contender for identifying DM production. This mechanism predicts that the Higgs boson decays into two DM particles. Thus, measuring the invisible Higgs decay branching fraction, denoted as $\text{BR}(H \rightarrow \text{inv})$, is necessary to test this framework.

This project aimed to place a more constraining upper limit on the branching fraction in the $t\bar{t}H$ channel, which CMS observed to be 43% [5]. A Particle Transformer was deployed as a global event-level classifier: trained and tested on simulated CMS Run 2 data. This architecture exploits pairwise jet kinematics and event-level correlations, unavailable to the object-level tagging and cut-based categorisation methods employed in [5]. The resulting signal-background separation was used to extract an expected upper limit on $\text{BR}(H \rightarrow \text{inv})$, this investigating the potential of event-level ML to advance sensitivity beyond existing $t\bar{t}H$ analyses.

2 Background

2.1 Dark Matter

Dark matter is a hypothetical form of matter that does not emit, absorb, or reflect light, but interacts gravitationally with visible matter and radiation. This concept was first hypothesised in 1933 by Fritz Zwicky to explain observations of galaxy clusters moving too fast to be gravitationally bound by their visible content [1].

It was Rubin and Ford’s systematic study of 60+ spiral galaxies (1970), however, that provided the first large-scale set of reproducible observational evidence [2]. They found that the outer regions of galaxies rotate far too quickly to be explained by visible matter alone: measurements pointed towards galaxies being surrounded by large invisible halos of mass.

The most concrete line of evidence came with the observation of the Bullet Cluster collision, where x-ray observations of hot gas (baryonic matter) revealed that visible matter interacted at the collision point, whilst gravitational lensing measurements indicated that significant gravitational mass passed through unaffected [3]. With this, the potential explanation of modified gravity (MOND) [6] was disfavoured as a complete alternative, lending strong observational support to dark matter as a fundamental constituent of the universe: comprising approximately 26.5% of its mass-energy budget and playing an essential role in structure formation across all cosmic scales [7].

2.2 The Higgs Boson and the Higgs Portal Mechanism

In 2012, the CMS and ATLAS collaborations observed the Higgs boson [8, 9]: an observable particle of ~ 125 GeV mass that arises due to excitations in the Higgs field (which was first theorised in 1964 [10–12]). This field permeates the universe, and is unique in being the only fundamental scalar field in the standard model (SM). Notably, this field couples to all massive SM particles to grant them their mass, with the coupling strength proportional to particle mass.

DM is observed to be non-relativistic and clustered on galactic scales, which requires non-zero rest mass. It is therefore natural to consider that DM particles inherit their mass through the same Higgs coupling mechanism [4]. Due to the coupling with the Higgs field, the Higgs boson would decay into a pair of DM particles provided that the kinematic threshold $m_H \geq 2m_{\text{DM}}$ is satisfied¹ [4]. This is known as the Higgs Portal Mechanism, and it functions as a DM detection framework since invisible Higgs decays can be measured: the SM predicts that $\text{BR}(H \rightarrow \text{inv}) = 0.1\%$, arising from the decay $H \rightarrow ZZ^* \rightarrow \nu\bar{\nu}\nu\bar{\nu}$ [13]. By comparison, experimental searches place upper limits of $\text{BR}(H \rightarrow \text{inv}) < 10.7\%$ at the 95% confidence level [14], allowing significant room for Beyond Standard Model (BSM) Higgs portal dark matter signatures.

2.3 Invisible Higgs Decays and the $t\bar{t}H$ Channel

The current observed limit for the $t\bar{t}H$ channel is $\text{BR}(H \rightarrow \text{inv}) < 43\%$ [5], sitting notably higher than the combined-channel limit of 10.7%. Amongst individual Higgs production modes, vector boson fusion (VBF) is the most constrained channel (14.5%), as its characteristic dijet topology and final state kinematics are highly distinctive [14]. In contrast, the $t\bar{t}H$ channel’s signal-to-background ratio of approximately 1:2000 pre-cuts has left its limit notably weaker.

The final state of $t\bar{t}H$ decay sees each top quark decay into a W and b -jet — with the W s in turn decaying either hadronically ($W \rightarrow q\bar{q}'$) or leptonically ($W \rightarrow \ell\nu$), providing a rich per-particle feature landscape. The study which placed the $t\bar{t}H$ upper limit of 43% used cut-based event categorisation; object-level Deep Neural Networks (DNNs) to tag individual particles such as b -jets, top-jets, and W -jets; and hadronic recoil as the main discriminating variable [5]: thus not exploiting event-level pairwise kinematic information. Therefore, this project identified the $t\bar{t}H$ channel as standing to gain the most from the introduction of event-level machine learning methods via particle transformer architecture.

¹By conservation of energy-momentum. The threshold condition requires the Higgs rest-mass energy to be sufficient to produce two DM particles at rest in the centre-of-mass frame.

2.4 The Particle Transformer

The particle transformer was first developed in 2022 for jet tagging² [15]. It is a permutation-invariant³ adaptation of the transformer design (2017) [16], which essentially extends the transformer’s self-attention mechanism to consider pairwise relationships between particles. In this case, this would allow the model to learn the specific kinematic correlations across final-state objects without explicit feature engineering.

This self-attention mechanism is the key mechanism underpinning transformers. Throughout training, the model uses it to learn to identify related particles and assign larger weights to those which are most relevant to understanding the global event. For example, this would allow the model to identify which particles likely originate from the same W boson decay. This project used the self-attention mechanism to capture and exploit event-level kinematic information: offering an avenue for improvement on the upper limit of 43% established in [5].

This model was deployed as an event classifier: each event was scored probabilistically on its likelihood of belonging to a certain event class. A score threshold was then chosen in order to separate signal from background as well as possible, and the separation was then fed into a statistical function (see Experimental Methods) to extract a limit.

3 Experimental Methods

3.1 Dataset, Simulation, Backgrounds

Supervised training of the neural network requires labelled data. Therefore, simulated datasets were used for training, following the CMS standard production procedures described in [5]. Monte Carlo events were generated using POWHEG for the matrix element calculation, interfaced with Pythia for parton showering, hadronisation, and underlying event modelling. Detector simulation was performed using GEANT4 to model the CMS detector response. These simulations emulated LHC Run 2 between 2016 and 2018. These datasets were provided by project supervisor Henning Flaecher and were not generated by the author.

The signal process was $t\bar{t}H$ with $H \rightarrow \text{inv}$ (decay described in §2.3). The following background classes were considered due to being strong sources of missing transverse energy (MET)⁴.

The first class was comprised of $t\bar{t}$ and single top quarks. These top quarks decay to a W boson and a bottom quark. When the W boson decays leptonically, the resulting neutrino escapes detection: producing genuine and irreducible

Class	Raw Events	σ_{vis} (fb)	$\delta\sigma_{\text{vis}}$
$t\bar{t}H$	196,068	10.03	0.02
$t\bar{t}$ & single tops	502,140	287.9	0.5
V +jets & EWK	128,340	85.2	0.4
QCD	1,009	11	1
Signal	196,068	10.03	0.02
Total background	631,489	384.1	1.2

Table 1: Visible cross sections σ_{vis} and MC statistical uncertainties $\delta\sigma_{\text{vis}}$ for each simulated event class, after requiring $n_{\text{jet}} \geq 5$, $n_{b\text{-jet}} \geq 1$, and $E_T^{\text{miss}} > 200$ GeV. Table does not include the QCD-enriched control region. Samples are normalised to the CMS Run 2 integrated luminosity of 138 fb^{-1} at $\sqrt{s} = 13$ TeV, with per-year luminosity weights applied separately for 2016–2018. $\delta\sigma_{\text{vis}} = \sqrt{\sum_i w_i^2} / \mathcal{L}$ reflects MC statistical uncertainty only.

MET. In cases of detector inefficiency or geometric acceptance, the lepton may also escape and thus lead to additional fake MET⁵. Semi-leptonic decay of the $t\bar{t}$ system leads to a final state containing MET, b-jets, and additional jets from hadronic decay of the second W : closely mimicking the signal. Similarly, the single top decays consist of the s-channel decay (tq'), t-channel ($tq'b'$), and tW channel (tW), where the leptonic decay of at least 1 W in each case results in jets, at least one b-jet, and MET.

The next class considered included V + jets and Electroweak decays (EWK). The former is produced when an initial-state quark radiates a W or Z boson — with additional jets from Quantum Chromodynamic (QCD) radiation — whilst the latter are produced via t-channel weak boson exchange [17]. Despite differing production mechanisms, both yield identical final states for the purposes of this analysis. This background also results in MET: leptonic decays of $W \rightarrow \ell\nu$ as per the above, $Z \rightarrow 2\ell$, or $Z \rightarrow 2\nu$ all result in either real or fake (if a lepton is lost to detector inefficiency or geometric acceptance) MET. These are always accompanied by jets, thus mimicking the signal when a jet is mistagged as a b-jet.

The last background class consists of QCD multijet radiation, in which jet mismeasurement produces fake MET. Due to the unreliability of Monte Carlo modelling of QCD, a dedicated QCD-enriched control region with less stringent cuts was included to improve the MC-based estimate of QCD contamination in the signal region and provide a more representative QCD sample for training. Thus, the post-cut signal region contains far fewer QCD events than the control region, as expected from stricter selection (seen when comparing Table 1 and Table 2).

The cross-sections of all classes in Table 1 demonstrate the crux of the problem behind the search in the $t\bar{t}H$ channel: signal is drowned out by background processes. The cross section values in this table are obtained by inverting the yield formula $N = \sigma_{\text{vis}} \cdot \mathcal{L}$, using the expected event yield N from simulations (after cuts).

Whilst the angular separation ($\Delta\phi$) between MET and

²Classifying individual jets rather than full events

³I.e. the order of input particles does not affect output.

⁴In High Energy Physics (HEP), this is defined as the momentum imbalance in the transverse plane (perpendicular to the proton beam’s axis). This is inferred from conservation of momentum, and is the signature of otherwise invisible particles.

⁵HEP terminology for MET caused purely by detector mismeasurement or particle misidentification.

leading jets exists as a kinematic discriminant between the signal and the first two classes, it is insufficient for any hard-cut approach. Feature distributions overlap significantly (see Figure 1), and the small signal-to-background ratio means that any cut on $\Delta\phi$ would accept overwhelming numbers of background events regardless of threshold selection. Therefore, event geometry ($\Delta\phi$) becomes a sanity check on the model’s behaviour (explored in §4.1.6).

3.2 Preprocessing

In order to preliminarily reduce background, the dataset was cut to only keep events that had at least: 5 cleaned jets⁶ present; 1 cleaned b-jet; and $E_T^{\text{miss}} > 200$ GeV. Requiring at least 5 jets selects the semi- and fully-hadronic $t\bar{t}$ topologies whilst rejecting fully leptonic decays, and strongly suppresses V+jets, EWK, and single-top backgrounds whose leading-order processes produce far fewer jets. The loose b-jet requirement (rather than needing both) accounts for potential b-jet losses due to detector inefficiency or geometric acceptance. Lastly, the $t\bar{t}H$ decay results in a boosted Higgs boson, making $E_T^{\text{miss}} > 200$ GeV a well-motivated selection threshold.

The neural network could take up to 10 jets as inputs per event: those with more than 10 jets were truncated to the 10 with highest p_T . Each jet was represented by 6 features:

- Transverse momentum (p_T): momentum perpendicular to beam axis.
- Pseudorapidity (η): angular coordinate along the beam direction, defined as $\eta = -\ln(\tan(\theta/2))$ where θ is the polar angle.
- Azimuthal angle (ϕ): angular coordinate around the beam axis.
- Invariant mass (M): derived from the four-momentum of the jet.
- Jet area: geometric measure of the jet’s angular extent.
- b-tagging discriminant: neural network output quantifying the likelihood that the jet originates from a b-quark.

Transverse momentum and invariant mass were log-transformed — using the function $\log_{10}(x+1)$ — to compress their dynamic range and improve numerical stability during training. The overlapping distributions for these features can be seen in Figure 1. The model was also given the magnitude and azimuthal angle ϕ of the MET vector as event-level features.

All features were normalised to zero mean and unit variance using weighted statistics computed on the training set, excluding masked entries. This normalisation is standard practice in machine learning, as features with larger

⁶A cleaned jet is a jet from which identified muons or electrons have been removed before jet energy computation to avoid poor calorimetric measurement of those particles [18].

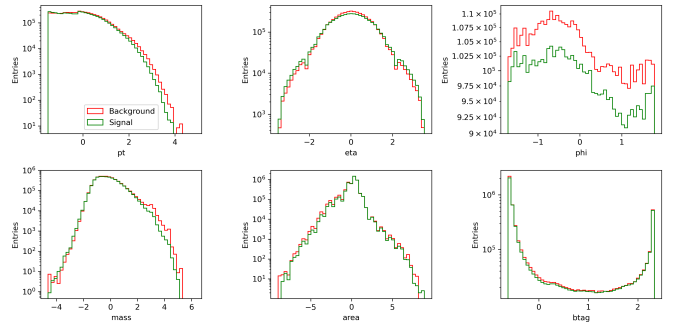


Figure 1: Normalised distributions of kinematic variables for signal ($t\bar{t}H \rightarrow \text{inv}$, green) and background (red) from each of the leading 10 jets of all post-cut events. The variables shown are transverse momentum (p_T); pseudorapidity (η); azimuthal angle (ϕ); invariant mass (M); jet area; and b-tagging discriminant. All distributions are normalised and demonstrate significant overlap between signal and background, illustrating the challenge of event discrimination using simple kinematic cuts.

magnitude would otherwise dominate gradient updates during training.

One-hot encoding⁷ was applied to label each event with its respective Run 2 data-taking period (2016A, 2016B, 2017, 2018). Since detector configurations and conditions varied across these periods, this information was retained as an auxiliary input, and concatenated with the learned event representation at a later stage (see §3.3) to allow the model to learn and account for systematic differences between periods whilst training on the larger combined dataset.

Lastly, the dataset was evenly split into two sets: a training set and a test set. To address the class imbalance during training, events were reweighted such that signal and each background class were equally represented in the dataset. This is important so that signal and background contribute equally to the loss function⁸, preventing the model from being biased toward predicting the majority class.

3.3 Model Architecture

Skeleton code for the model was provided at the start of the project. The architecture consists of three main stages. First, event features are processed through an embedding layer that learnt a new 64-dimensional representation via non-linear transformations. This is used to emphasise the most relevant features for signal-background separation. These vectors are then passed on through 8 sequential processing blocks, performing multi-head self-attention⁹

⁷One-hot encoding refers to converting categorical data into vectors. For example, run-years are encoded as (2016a, 2016b, 2017, 2018)^T, with the vector (0, 0, 1, 0)^T signifying an event from 2017.

⁸A loss function is a mathematical measure of how wrong the model’s predictions are. During training, the model’s parameters are updated iteratively to minimise this quantity via gradient descent.

⁹Multi-head attention refers to running several attention operations in parallel, with each attending to different aspects of the event. One may learn spatial relationships whilst another learns momentum correlations, for example. The output of this block is then con-

followed by further non-linear transformations. These 8 encoder blocks allow the model to iteratively refine its understanding of the event structure, progressively building more complex patterns from simple jet relationships.

After this, the code concatenates the period-specific one-hot vector encoding with each event’s learned vector representation. This is deliberately placed here (rather than at the input level) so that the model can first learn relationships within each event, and then correct for period-specific systematic differences.

This combined representation is then passed through 3 classification blocks — again using attention — followed by a 3-layer output network. The final output layer produces a probability between 0 and 1 per class. This is done by mapping the output value from the model — which can be any real number — onto the sigmoid activation function (for binary classifiers):

$$\sigma(x) = \frac{1}{1 + e^{-x}}, \quad (1)$$

or the softmax function (for multiclassifiers):

$$\text{softmax}(x_i) = \frac{e^{x_i}}{\sum_j^{\text{all classes}} e^{x_j}}. \quad (2)$$

This probability represents the model’s confidence that an event belongs to a certain class. A 1 represents complete certainty that it belongs to a class, and a 0 represents certainty that it does not belong, whilst intermediate scores represent degrees of uncertainty in the classification.

The complete architecture contained 637k trainable parameters. Two classifier configurations were evaluated in this project: a binary classifier (where background classes were grouped into one), and a multiclassifier where the model was trained to distinguish between background classes. Results are compared in §4.3.

3.4 Training Procedure

Training took place on the training dataset, and included both the signal region and the QCD-enriched control region (whilst later testing computed exclusively on the signal region). Visible cross sections seen in the training data are shown in Table 2. This set was further divided into 80% training data and 20% validation data. The model was thus trained purely on training data, and validation data was used to monitor out-of-sample loss and prevent overfitting. If the model’s performance on the validation set (measured by loss) did not improve by a threshold value of 10^{-4} for a consecutive streak of 20 epochs¹⁰, training was automatically stopped. If there was no early stopping, the model was set to run 200 epochs.

Class	Raw Events	σ_{vis} (fb)	$\delta\sigma_{\text{vis}}$
$t\bar{t}H$	303,563	15.68	0.03
$t\bar{t}$ & single tops	1,081,518	683.1	0.7
V +jets & EWK	266,979	154.5	0.6
QCD	62,991	587	8
Signal	303,563	15.68	0.03
Total background	1,411,488	1,424.6	8.1

Table 2: Visible cross sections σ_{vis} and MC statistical uncertainties $\delta\sigma_{\text{vis}}$ for each simulated event class, after requiring $n_{\text{jet}} \geq 5$, $n_{b\text{-jet}} \geq 1$, and $E_T^{\text{miss}} > 200$ GeV. Table includes the QCD-enriched control region. Samples are normalised to the CMS Run 2 integrated luminosity of 138 fb^{-1} at $\sqrt{s} = 13$ TeV, with per-year luminosity weights applied separately for 2016–2018. $\delta\sigma_{\text{vis}} = \sqrt{\sum_i w_i^2} / \mathcal{L}$ reflects MC statistical uncertainty only.

The aim was to minimise cross-entropy loss — equivalent to maximum likelihood estimation for categorical distributions — which penalises confident incorrect predictions heaviest [19]. Mathematically:

$$\mathcal{L}_{\text{cross entropy}}^{\text{per event}} = -\log(\hat{y}_c), \quad (3)$$

where $\hat{y}_c$ is the model’s predicted probability for the event’s true class. This was combined with a distance correlation (DisCo) loss term [20] to prevent the model from relying on MET patterns. If statistical dependence between MET and score were not penalised, Monte Carlo mismodelling of MET could corrupt the network’s transferability to real data, making findings untrustworthy. This loss term is described by:

$$\mathcal{L}_{\text{DisCo}} = d\text{Corr}^2(s, E_T^{\text{miss}}) \quad (4)$$

$$\mathcal{L}_{\text{DisCo}} = \frac{d\text{Cov}^2(s, E_T^{\text{miss}})}{d\text{Var}(s) \cdot d\text{Var}(E_T^{\text{miss}})}, \quad (5)$$

where s is the model’s output score and $d\text{Cov}$ is the distance covariance. These loss terms are combined linearly with the cross-entropy term such that total loss is described as:

$$\mathcal{L} = \mathcal{L}_{\text{cross entropy}} + \lambda \times \mathcal{L}_{\text{DisCo}}. \quad (6)$$

This λ was used as a scheduling factor — increasing exponentially from 0.005 to 0.2 as training progressed — so that the model may first learn effective classification before the decorrelation constraint is enforced.

Model weights were updated using the AdamW optimiser [21] with an initial learning rate of 0.0004 and batch size of 1024. The learning rate controls how quickly the model adjusts its weights; smaller values lead to increased stability at the cost of slower learning. This is reduced by a factor of 10 if validation performance does not improve for 15 consecutive epochs, with a minimum learning rate set at 10^{-6} to ensure continued training progress. Weight decay (L2 regularisation) was set at 0.01 to penalise overly complex weight values, encouraging simpler solutions that generalise better to unseen data.

catenated and projected, giving the model richer representational capacity than a single attention operation.

¹⁰One epoch refers to one complete pass through the entire training dataset. Models are typically trained for many epochs, with parameters updated incrementally via mini-batches of data within each epoch.

3.5 Validation and Performance Metrics

Model performance was evaluated out-of-sample, on the test set. The following metrics were used to assess both overall discrimination power and physics sensitivity:

- **Area under the curve (AUC), for a receiver operating characteristic (ROC) curve:** Measures the model’s ability to distinguish signal from background across all decision thresholds, ranging from 0.5 (random) to 1 (perfect). A model scoring an AUC below 0.5 is simply inverted, where decisions should be made opposite to the model’s prediction. An AUC of 0 thus corresponds to a perfectly inverted model.
- **Output score distributions:** Visualises the model’s predicted probability distributions for signal and background events, directly showing the degree of separation achieved. For a binary classifier, scores scale from 0 (background-like) to 1 (signal-like).

In the multiclass case, each class is scored on a probability scale of 0 – 1, with each event’s class-wide probabilities summing to 1. To quantify signal-background separation, the score is expressed as a log-likelihood ratio:

$$s = \log \left[\frac{P(t\bar{t}H)}{\sum(\text{bkg} + \epsilon)} \right] = \log \left[\frac{P(t\bar{t}H)}{1 - P(t\bar{t}H) + \epsilon} \right], \quad (7)$$

with ϵ being an infinitesimally small number used to guard against division by zero. This function maps the score on $(-\infty, \infty)$, with positive scores indicating signal-likeness.

- **Confusion matrix:** Assigns each event to the class with the highest predicted probability, reporting the fraction of true class i events predicted as class j . This summarises class-wise assignment accuracy.
- **Distance Correlation (DisCo):** Quantifies statistical independence between model output score and MET, reported alongside Pearson correlation coefficient (r) for comparison. DisCo and Pearson’s r both range from 0 to 1, where 0 means complete statistical independence and 1 means perfect dependence. DisCo captures non-linear correlations whilst Pearson’s r captures linear correlations. This project aims to minimise DisCo in order to prevent MET mismodelling bias.
- **Expected significance (Z_{exp}):** Calculated using Profile Likelihood Significance with Background Uncertainty [22], assuming observed event counts match their expected values. Significance was evaluated at three background uncertainty scenarios ($\sigma_b = 5\%, 10\%, 20\%$) to assess sensitivity across realistic systematic variations. The Monte Carlo samples (generated at $\text{BR}(H \rightarrow \text{inv}) = 100\%$) were downsampled in order to match the branching fraction desired for statistical testing. This ensures the analysis probes a physically realistic beyond-Standard-Model scenario (§3.6).

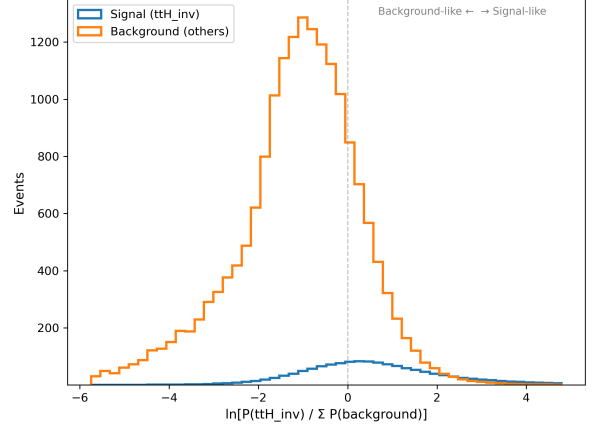


Figure 2: Multiclassifier log-likelihood ratio distribution on the complete test set. Event counts scaled to cross sections, and $t\bar{t}H$ event counts scaled according to $\text{BR}(H \rightarrow \text{inv}) = 10\%$.

3.6 Statistical Limit Extraction

The limit extraction procedure began by binning event log-likelihood scores into 50 bins, with the signal event yields downsampled to match the branching fraction for which the model’s sensitivity is being tested¹¹.

Each bin edge served as a candidate score threshold, where all events above this would be selected as part of the signal region. Expected signal (s) and background ($b\sigma_b$) yields in this region were used to compute Z_{exp} via the Profile Likelihood formula [22]:

$$Z_{\text{exp}} = \sqrt{2 \left[(s+b) \ln \frac{(s+b)(b+\sigma_b^2)}{b^2 + (s+b)\sigma_b^2} - \frac{b^2}{\sigma_b^2} \ln \left(1 + \frac{s\sigma_b^2}{b(b+\sigma_b^2)} \right) \right]} \quad (8)$$

where $\sigma_b = (5\%, 10\%, 20\%)$ parametrises flat background yield uncertainty scenarios. Z_{exp} quantifies the hypothesised signal’s deviation from background-only expectations in units of standard deviations (σ).

4 Results

This section will present results from the multiclassifier and binary classifier versions of the model. Focus will be placed on the multiclassifier, as its per-class probability outputs allow verification of whether the model has learnt the physically expected kinematic structure of each event class. The binary classifier serves as a methodological baseline.

4.1 Multiclassifier Performance

4.1.1 Score Distribution Plots

Figure 2 demonstrates clear signal-background separation, although suffering from the physical class imbalance. Signal peaks at $s \simeq +0.25$, whilst backgrounds peak at $s \simeq -1.25$.

¹¹This analysis uses expected counts on simulated data throughout; results are therefore projected sensitivities rather than observed limits.

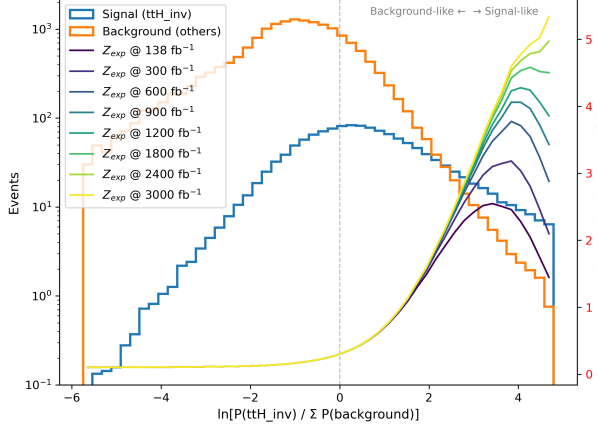


Figure 3: Multiclassifier log-likelihood ratio distribution on the complete test set. Event counts scaled to cross sections, and ttH event counts scaled according to $\text{BR}(H \rightarrow \text{inv}) = 40\%$. Expected significances assume a background error of $\sigma_b = 10\%$ at different Large Hadron Collider luminosities. Left logarithmic axis for the histogram, Right axis for the expected significance curves.

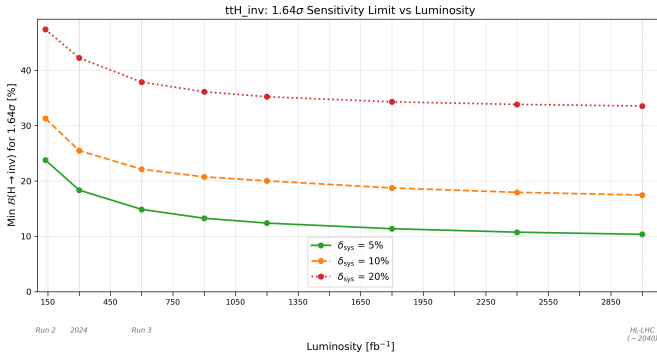


Figure 4: Minimum upper limit ($H \rightarrow \text{inv}$) at a 95% confidence level vs LHC integrated luminosity, for $\sigma_b = 5, 10, 20\%$, using a multiclassifier.

Additionally, Figure 3 demonstrates the model's projected sensitivity at $\text{BR}(H \rightarrow \text{inv}) = 40\%$. For a luminosity of 138fb^{-1} (integrated luminosity achieved by CMS Run 2) and background error of $\sigma_b = 10\%$, the optimal threshold yielded an expected significance of 2.55σ . Even when luminosity is increased tenfold (as is the aim of the LHC by 2040, denoted HL-LHC for High Luminosity Large Hadron Collider), the continued separation between the curves suggests that statistical uncertainties remain dominant over systematics (within the simplified background uncertainty model adopted here).

4.1.2 Projected Upper Limit Sensitivity

Figure 4 shows the projected 95% CL upper limit sensitivity. For a background yield error $\sigma_b = 10\%$, the model achieves a 31.3% upper limit at Run 2 luminosity (138fb^{-1}) and 17.5% at HL-LHC (3ab^{-1}). The rate of gain in sensitivity for larger luminosity decreases as background yield error rises.

Figure 5 further shows the projected $\text{BR}(H \rightarrow \text{inv})$ up-

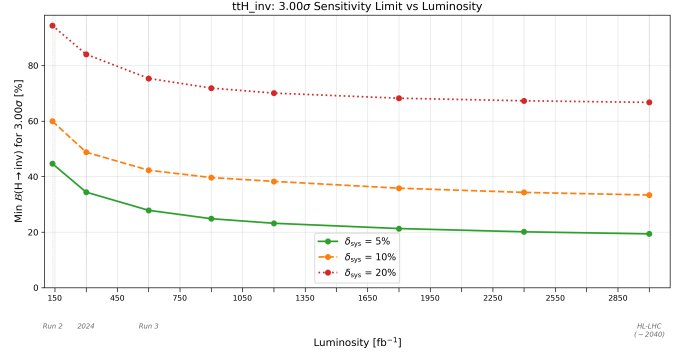


Figure 5: Minimum upper limit ($H \rightarrow \text{inv}$) branching fraction sensitivity at 3σ discovery significance vs LHC integrated luminosity, for $\sigma_b = 5, 10, 20\%$, using a multiclassifier.

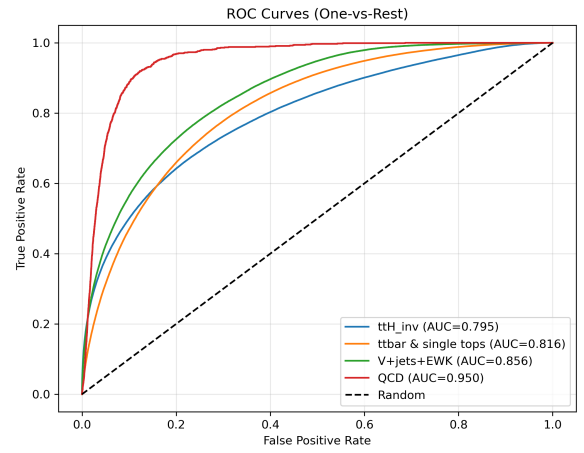


Figure 6: One vs Rest ROC curves for test set. This scores each class against the remaining classes. For a continuous range of thresholds on the model score outputs, the y-axis measures the percentage of class events correctly identified, whilst the x-axis measures the percentage of non-class events falsely flagged as class. AUC values shown per-class in legend.

per limit sensitivity for a 3σ discovery significance. For a background baseline error $\sigma_b = 10\%$, the model achieves a 60.0% upper limit at Run 2 luminosity (138fb^{-1}) and 33.4% at HL-LHC (3ab^{-1}).

4.1.3 ROC curve

Figure 6 shows the model yielded an AUC of ttH : 0.795; $t\bar{t}$ & single tops: 0.816; V +jets & EWK: 0.856; QCD: 0.950. These AUCs represent the probability that the model correctly ranks a random event of this class above a random non-class event. The ordering of these AUC strengths follows the expected physics: ttH yielded the lowest AUC as the background classes were selected precisely because they share similar final states with the signal; $t\bar{t}$ & single tops are the closest background to ttH , so $t\bar{t}$ & single tops consequently yield low AUC due to these similar backgrounds. V +jets & EWK are slightly more physically distinct, so AUC improves; and lastly QCD shares no kinematic similarities with the previous classes, and thus yielded a very high AUC. The ttH AUC of 0.795 quantifies the model's discrimination power, and serves as

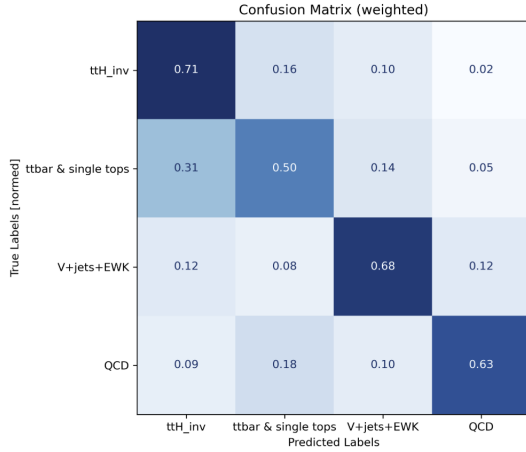


Figure 7: Confusion matrix for true labels (rows, y-axis) vs predicted labels (columns, x-axis) by highest class probability. Cell (i, j) : fraction of true class i events predicted as class j (0 =never, 1 =always).

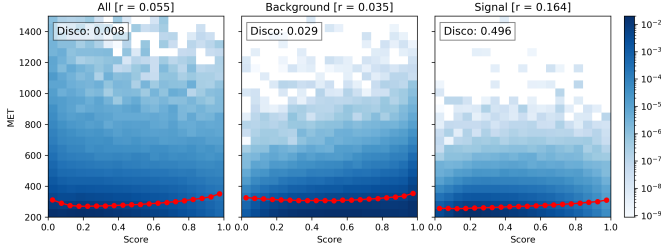


Figure 8: 2D heatmaps of MET (y-axis) and model score for the true event class (x-axis) on the full test set, background classes only, and signal class only. Each plot generated with 40,000 sample events. Red lines show average MET per score bin division.

the primary classifier performance metric against which the binary baseline is compared in Table 3.

4.1.4 Confusion Matrix

Figure 7 visualises class prediction performance via the confusion matrix. The model correctly identifies $t\bar{t}H$ events with probability 0.71 — its strongest performance in this matrix. Physical expectations hold, as the model is most confused by the $t\bar{t}H$ and $t\bar{t}$ & single-top pairing (due to similar jet topology), while V +jets/EWK and QCD distribute misclassification more evenly across other classes.

4.1.5 MET Decorrelation

Figure 8 shows how well the model decorrelated results with MET: a key goal in ensuring results are safeguarded from potential Monte Carlo MET mismodelling. The overall test set had a Pearson’s correlation coefficient¹² $r = 0.055$, whilst background and signal yielded coefficients of $r = 0.035$ and $r = 0.164$ respectively. This indicates that MET and model score (defined as the predicted probability of each event’s true class) were very weakly linearly correlated. DisCo for the complete test set was 0.008,

¹²Pearson’s r measures linear correlation, ranging from -1 (perfect negative) to $+1$ (perfect positive), with $r = 0$ indicating no linear dependence.

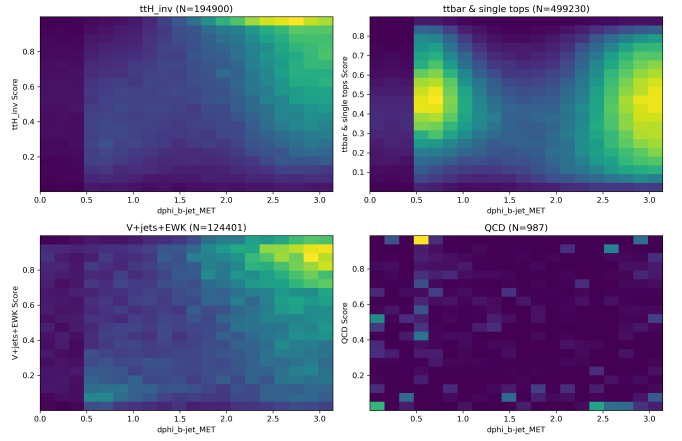


Figure 9: 2D heatmaps of per-class model score (y-axis) vs $\Delta\phi(\text{MET}, \text{leading b-jet})$ (x-axis) for each true class. Left-column suppression reflects simulation-level cuts requiring minimum $\Delta\phi > 0.5$ between four leading jets and MET.

sitting significantly lower than the individual subgroup’s DisCo of 0.029 and 0.496 for background and signal respectively. This is because background and signal DisCo are computed against their respective class probabilities, which occupy separated score regions. Pooling them conflates two different score axes, such that the aggregate DisCo is not a simple combination of the subgroup values. $t\bar{t}H$ events naturally have higher MET from the true invisible decay — so some positive score-MET correlation is physically expected — whilst background events may score highly in their own classification without necessarily having high MET. The low physical weight of $t\bar{t}H$ events further reduces their contribution to the combined DisCo, causing the aggregate to be dominated by the near-zero background correlation.

4.1.6 Feature Analyses

Lastly, an analysis of how per-class scoring was related to physical features was conducted, testing whether the model learnt physically meaningful relationships.

Figure 9 investigates the relationship between per-class event scoring and the angular separation between the event’s b-jet and the lead b-jet — defined as the highest p_T b-jet, with a b-tagging working point¹³ $\text{Jet_btagDeepFlav} > 0.25$ to balance mistag rate and efficiency [23]. These plots demonstrate that the model learnt the expected physics correctly: in $t\bar{t}H$, the Higgs recoils against the $t\bar{t}$ system as a whole, so MET from $H \rightarrow \text{inv}$ is directed away from the entire visible final state rather than tied to any single decay product. The top quarks have their own angular separation before decaying into b-jets and W bosons, leaving the event’s MET at a medium-to-large $\Delta\phi$ from the b-jets.

In the $t\bar{t}$ case, the neutrino MET is tied to the lep-

¹³A b-tagging working point is a threshold on the b-tagging discriminant score above which a jet is classified as a b-jet. Looser thresholds increase efficiency (more true b-jets accepted) at the cost of higher mistag rates (more light-flavour jets falsely accepted), and vice versa.

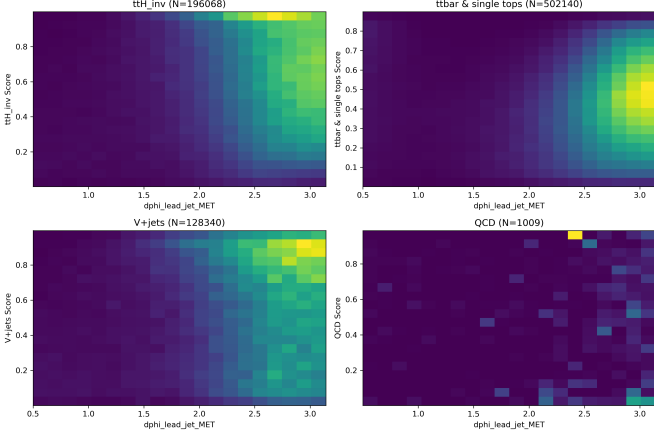


Figure 10: 2D heatmaps of per-class model score (y-axis) vs $\Delta\phi$ (MET, leading jet) (x-axis) for each true class. Left-column suppression reflects simulation-level cuts requiring minimum $\Delta\phi > 0.5$ between four leading jets and MET.

tonic top decay $W \rightarrow \ell\nu$, meaning MET is either roughly aligned or anti-aligned with the lead b-jet — resulting in either a small or large $\Delta\phi$. This logic is equally applicable to the single tops decay. Thus, the model appropriately learnt this physical feature for this class — with reduced certainty as the distributions cluster at scores of 0.5.

For V +jets & EWK, the neutrino comes from the V and tends to be produced opposite the jets, which are predominantly mistagged light-flavour jets rather than true b-jets: thus, a large $\Delta\phi$ is expected and well-learnt by the model. Lastly, QCD $\Delta\phi$ is correctly identified as showing no physically meaningful structure, as their MET arises purely from detector inefficiency rather than true invisible particles.

Figure 10 extends the analysis from Figure 9 to the angular separation between MET and the lead jet (rather than lead b-jet). The $t\bar{t}H$ heatmap closely resembles the result in Figure 9 as expected: the leading jet originates from the same top-decay system as the leading b-jet, preserving the same geometry.

Given that selection cuts required $n_{\text{cleanedjets}} > 5$ and $E_T^{\text{miss}} > 200\text{GeV}$, the most common surviving decay channel in the $t\bar{t}$ & single tops class was the semi-leptonic $t\bar{t}$ decay, in which the leptonic W ($W \rightarrow \ell\nu$) decays in one hemisphere and hadronic W ($W \rightarrow q\bar{q}'$) in the other. The hadronic side contributes more visible jet activity than the leptonic side (which produces just one b-jet), meaning that the leading jet is most often found on the hadronic side. Beyond this probabilistic explanation, it is also possible that jets in the narrow region resulting in the hadronic hemisphere overlap, thus artificially increasing p_T through mismeasurement: further increasing the likelihood of finding the lead jet in the opposite hemisphere to the neutrino. For single top decays, the tW mode dominates post-cuts. By similar logic, cuts lead to a majority of semi-leptonic events, where the leptonic W decay’s MET is back-to-back with the hadronic W decay’s jets. Both of these cases leave the MET roughly anti-aligned with the lead jet, as supported by the model in Fig 10.

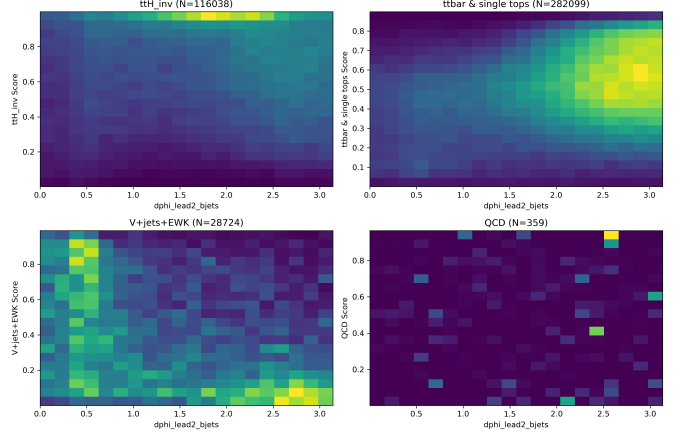


Figure 11: 2D heatmaps of per-class model score (y-axis) vs $\Delta\phi$ (leading 2 b-jets) (x-axis) for each true class.

Lastly, the V +jets & EWK events generate neutrinos from the leptonic decay of their V boson. Since the boson is produced back-to-back with jets, the MET is anti-aligned with the lead jet. The model learnt to identify this with confidence, seen by the high-scoring cluster in the large $\Delta\phi$ region. Again, the QCD class follows expectations of no physical correlations due to the MET resulting from detector inefficiency or geometric acceptance.

Figure 11 investigates the relationship between per-class event scoring and the angular separation between the event’s two leading b-jets, demonstrating the model precisely learnt this element of expected event geometry. This analysis left out events that did not contain two b-jets clearing the working point requirement of $\text{Jet.btagDeepFlav} > 0.25$. The $t\bar{t}H$ case shows a heavy cluster of high-scoring events in the middle range of $\Delta\phi$, following the expected boosted $t\bar{t}$ kinematics. The model also identifies this feature successfully for the $t\bar{t}$ & single tops class, with clustering taking place at around $\Delta\phi = \pi\text{rad}$ with high scoring. This is expected from the back-to-back $t\bar{t}$ /t-channel single top decay topology. Lastly, both V +jets & EWK and QCD classes show irregular, non-physical patterns due to their b-jets originating from mistagging rather than true decays.

4.2 Binary Classifier Baseline Performance

4.2.1 Projected Upper Limit Sensitivity

Figure 12 shows the projected 95% CL upper limit sensitivity when using a binary classifier rather than a multiclassifier. For a background yield error $\sigma_b = 10\%$, the model achieves a 28.4% upper limit at Run 2 luminosity (138 fb^{-1}) and 16.4% at HL-LHC (3 ab^{-1}). As in Figure 4, the rate of gain in sensitivity for larger luminosity decreases as background yield error rises. Figure 13 shows the projected $\text{BR}(H \rightarrow \text{inv})$ upper limit sensitivity for a 3σ discovery significance, using a binary classifier. For a background baseline error $\sigma_b = 10\%$, the model achieves a 54.5% upper limit at Run 2 luminosity (138 fb^{-1}) and 30.6% at HL-LHC (3 ab^{-1}).

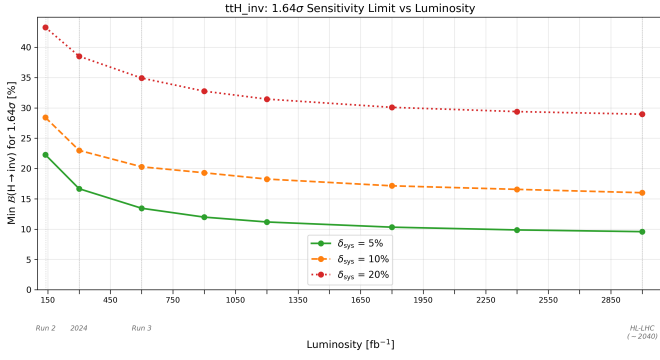


Figure 12: Minimum upper limit ($H \rightarrow \text{inv}$) at a 95% confidence level vs LHC integrated luminosity, for $\sigma_b = 5, 10, 20\%$, using a binary classifier.

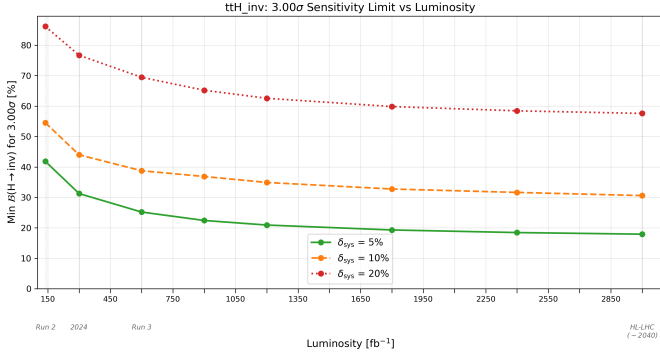


Figure 13: Minimum upper limit ($H \rightarrow \text{inv}$) at a 3σ discovery significance vs LHC integrated luminosity, for $\sigma_b = 5, 10, 20\%$, using a binary classifier.

4.2.2 ROC curve

Figure 14 shows the binary classifier yielded an AUC of 0.815. This means that in the binary classifier case, a random $t\bar{t}H$ event will be ranked higher than a random background event 81.5% of the time. The ROC curve rises steeply near the origin, with the true positive rate reaching ~ 0.675 at a false positive rate of 0.2, demonstrating concentrated discrimination in the high background rejection regime.

4.3 Classifier Performance Comparison Summary

Key performance metrics for both classifier configurations are summarised in Table 3.

The binary classifier outperforms the multiclassifier on

Metric	Multiclassifier	Binary Classifier
95% CL UL, 138 fb^{-1}	31.3%	28.4%
95% CL UL, 3 ab^{-1}	17.5%	16.4%
3σ UL, 138 fb^{-1}	60.0%	54.5%
3σ UL, 3 ab^{-1}	33.4%	30.6%
AUC	0.795	0.815

Table 3: Comparison of key metrics between binary and multiclassifier models. UL = Upper Limit on the $\text{BR}(H \rightarrow \text{inv})$. Upper limit values assume background yield errors of $\sigma_b = 10\%$.

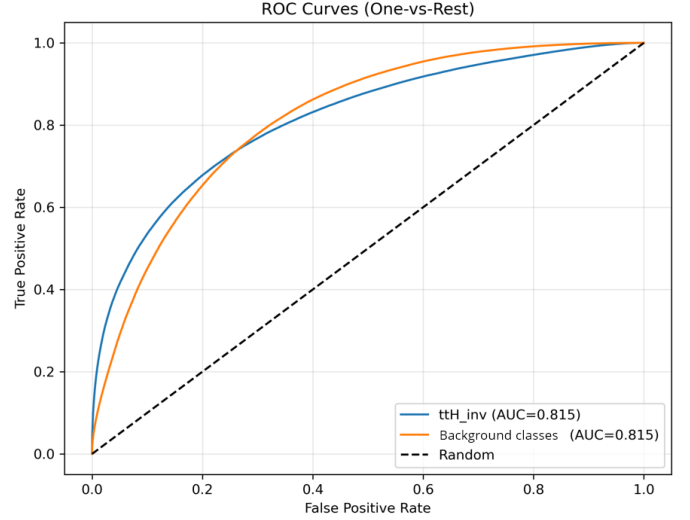


Figure 14: ROC curve for test set. This scores signal or background against the other. For a continuous range of thresholds on the model score outputs, the y-axis measures the percentage of class events correctly identified, whilst the x-axis measures the percentage of non-class events falsely flagged as class. AUCs shown in legend.

both raw discrimination power and limit sensitivity across all luminosity scenarios, with the interpretation of this result discussed in §5.1.

5 Discussion

5.1 Classifier Performance and Comparison

A key finding from this project is that a binary classifier outperforms a multiclassifier in separating $t\bar{t}H$ events from background. This is directly demonstrated in Table 3, with the binary classifier yielding upper limits on the branching fraction that are approximately 10% lower in relative terms than those obtained by the multiclassifier. Alongside this, the binary classifier's AUC of 0.815 exceeds the multiclassifier's 0.795 by an absolute margin of 0.02. This arises naturally from the binary classifier operating with a single decision boundary: allowing it to optimise signal-to-background separation directly whilst a multiclassifier distributes its representational capacity across multiple class boundaries. The multiclassifier must simultaneously learn inter-background boundaries — distinguishing $t\bar{t}$ from V +jets, for example — which do not directly optimise $t\bar{t}H$ signal sensitivity: thereby diluting its discriminative focus.

Nonetheless, a multiclassifier is preferred in the context of this project, despite the binary classifier's edge in physics sensitivity. From a validation standpoint, separating background classes grants the opportunity to explore the model's learnt physics (as done in §4.1.6). Confirming the model learnt expected features such as $\Delta\phi(\text{MET}, \text{lead b-jet})$; $\Delta\phi(\text{MET}, \text{lead jet})$; and $\Delta\phi(2 \text{ lead b-jets})$ provides confidence that the model's representations are well-motivated rather than exploiting Monte Carlo artefacts.

Therefore, if the objective is solely to achieve a stronger expected limit on the branching fraction, this project finds that a binary classifier is the preferred classification strategy for maximising expected sensitivity, although the out-performance is marginal. Repeated training runs did not change the qualitative conclusions or the reported metrics, suggesting the result is stable under the training procedure used. If interpretability is more important, the multiclassifier architecture allows deeper analysis of the model’s behaviour with regards to the underlying event-level physics.

5.2 Physical Validation

The multiclassifier model successfully learnt the expected kinematic correlations underpinning the simulated events, providing confidence that the model’s representations are well-motivated. Feature analyses in §4.1.6 found that the model learnt the expected $\Delta\phi$ features as expected with high certainty — except for the $t\bar{t}$ & single tops class. This class often received lower scores on these physical-feature clusters, reflecting the model’s reduced certainty due to kinematic overlap with both signal and the V +jets & EWK class (as seen in Figure 7). In Figure 6, AUC was ordered according to the expected physical separation difficulty: classes mimicking the signal closest were hardest to identify correctly (QCD > V +jets & EWK > $t\bar{t}$ & single tops > $t\bar{t}H$).

5.3 Comparison to Literature

The multiclassifier yielded an expected 95% confidence level upper limit of 31.3% at LHC Run 2 luminosity 138fb^{-1} and background uncertainty $\sigma_b = 10\%$, suggesting substantial improvement over the existing $t\bar{t}H$ channel limit of 43% [5]. This suggests that event-level ML classification has potential to provide additional discriminating power relative to the CMS approach described in §2.3.

As for the discovery-sensitivity regime of 3σ , the multiclassifier yielded a projected branching-fraction threshold of 42.3% (Figure 5) at a luminosity of 600fb^{-1} and a background uncertainty of $\sigma_b = 10\%$. This corresponds approximately to the integrated luminosity expected by the end of LHC Run 3.

5.4 Limitations and Future Work

However, a direct comparison with the CMS observed limit is not entirely meaningful: the CMS limit was extracted via a full shape-based maximum likelihood fit incorporating per-process systematic nuisance parameters for jet energy scale, b-tagging efficiency, and QCD normalisation [5], whilst this analysis employed a simplified flat background uncertainty parametrisation σ_b . Additionally, this upper limit was set using real LHC data, whereas this analysis was confined to producing an expected limit on simulation.

Two future developments would be needed in order to bridge this comparative gap. Firstly, the model devel-

oped in this project must be applied to real LHC data. Systematic differences — such as MET mismodelling and detector smearing — may arise between simulation and real data, thus motivating the need for the model to be robust to simulation-to-data transfer. Domain adaptation techniques, such as introducing gradient reversal layers, would encourage the model to explicitly learn features that stay invariant between simulation and real data. Additionally, a complete statistical analysis of uncertainty is necessary for comparison to CMS results: introducing the same full shape-based maximum likelihood fit incorporating per-process systematic nuisance parameters would make the statistical treatment of background error directly comparable to that of [5].

The $t\bar{t}H$ distance correlation of 0.496 represents a moderately strong non-linear dependence between signal event scores and the magnitude of MET. Therefore, the magnitude of MET provides the model with substantial information in determining how signal-like a given event is, despite the usage of a DisCo loss term (§3.4). This is physically expected and unavoidable to a large extent, as $H \rightarrow \text{inv}$ events have intrinsically high MET. For background events, the weak DisCo of 0.029 is a major success: since background events overwhelm signal, slight Monte Carlo mismodelling of MET has the potential to strongly corrupt expected background yields when transferring to real data. This low DisCo score confirms that this risk is well-controlled by this model.

In order to ensure maximal transferability to real data, a future improvement to this model would be to introduce class-wise DisCo loss. This project’s usage of DisCo loss suppressed MET-score correlations for all event types, regardless of its desirability per-class. Background classes should have these correlations suppressed in order to minimise Monte Carlo MET mismodelling bias, whilst suppressing signal correlation is undesirable as MET is physically expected to be large for $H \rightarrow \text{inv}$ events. To avoid this tension, introducing a class-specific DisCo loss term would allow aggressive MET-score decorrelation for backgrounds whilst avoiding unnecessarily constraining the signal’s genuine MET dependence.

6 Conclusions

This project succeeded in developing a Particle Transformer multiclassifier, with results suggesting the architecture’s event-level approach is a viable route for improvements to be made on the CMS upper limit on the $H \rightarrow \text{inv}$ branching fraction in the $t\bar{t}H$ channel. The model found the expected 95% confidence level upper limit to be 31.3% at the LHC Run 2 integrated luminosity of 138fb^{-1} using a flat background yield uncertainty of $\sigma_b = 10\%$, compared to the CMS upper limit of 43%.

It was also found that a binary classifier — where background classes were grouped into one — marginally outperformed the multiclassifier design. However, the interpretability of the multiclassifier allowed for deeper analysis of the model’s behaviour in response to the underly-

ing event-level physics. This interpretability was used to confirm the model learnt kinematic features used to distinguish events via score-feature heatmaps. Results were also successfully decorrelated from MET — with DisCo of 0.029 and 0.496 (moderate, as expected) for background and signal respectively — mitigating potential Monte Carlo MET mismodelling bias when transferring the model to real LHC data.

This project was an expected-only analysis at the simulation boundary. Progressing to a full observed result would require the introduction of real LHC data, and incorporating domain adaptation during training to allow the model to focus on features that exist in both simulated and real data. For complete statistical rigour and comparability with the CMS observed limit, it would be beneficial to extend the background error propagation from a flat background yield uncertainty to a full shape-based maximum likelihood fit incorporating per-process systematic nuisance parameters. Lastly, it was identified that the dataset-wide MET-score decorrelation lead to conflicts of interest between MET-heavy and MET-light classes. By introducing class-wise MET-score decorrelation, each class’ decorrelation can be optimised according to its physics, thus optimising transferability to real data.

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